

Input Data and Parameter Values

For completeness and reproducibility, all input parameters adopted in the computational experiments are summarised in Tables 1–5.

Table 1: Financial and temporal parameters

Parameter	Symbol	Value	Unit
Interest rate	i	0.05	–
Time step	Δt	1	h
Months in horizon	M	12	–
Hours per representative day	T	24	–
Days per month	N_m	[31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31]	days
Asset lifetimes	Y_k	15	years

Table 2: Technology-related parameters

Technology	Parameter	Symbol	Value	Unit
PV	Max capacity	$\Gamma_{\max}^{\text{PV}}$	2,000	kW
PV	Standard test irradiance	I^{STC}	1.00	kW/m ²
CHP	Electrical efficiency	η_{CHP}	0.30	–
CHP	Thermal loss factor	η_{Loss}	0.03	–
CHP	Fuel coefficient	ζ_1	3.33	–
CHP	Fixed fuel term (on)	ζ_2	40	kW
CHP/GB	H ₂ blending limit	κ_{H2}	0.20	–
Electrolyzer	Efficiency (LHV)	η_{EL}	0.67	–
Fuel cell	Electric efficiency	$\eta_{\text{HFC}}^{\text{elec}}$	0.40	–
Fuel cell	Heat efficiency	$\eta_{\text{HFC}}^{\text{heat}}$	0.45	–
Gas boiler	Efficiency	η_{GB}	0.92	–
Electric boiler	Efficiency	η_{EB}	0.98	–
Battery	Charge efficiency	$\eta_{\text{BT}}^{\text{ch}}$	0.95	–
Battery	Discharge efficiency	$\eta_{\text{BT}}^{\text{dis}}$	0.95	–
Hydrogen	LHV	LHV_{H2}	33.33	kWh/kg
Grid	electricity carbon-cost coefficient	ϕ^e	0.02400	€/kWh
Natural	gas carbon-cost coefficient	ϕ^g	0.01616	€/kWh

Table 3: Cost parameters.

Technology	Parameter	Symbol	Value	Unit
PV	Specific CAPEX	γ_{PV}	700	€/kW
CHP	Specific CAPEX	γ_{CHP}	1,000	€/kW
Electrolyzer	Specific CAPEX	γ_{EL}	2,200	€/kW
Fuel cell	Specific CAPEX	γ_{HFC}	2,500	€/kW
Gas boiler	Specific CAPEX	γ_{GB}	120	€/kW
Electric boiler	Specific CAPEX	γ_{EB}	80	€/kW
Battery	Specific CAPEX	γ_{BT}	120	€/kWh
H ₂ storage	Specific CAPEX	γ_{HST}	10.50	€/kWh
PV	Annual FO&M	ϵ_{PV}	10.50	€/(kW year)
CHP	Annual FO&M	ϵ_{CHP}	40.00	€/(kW year)
Electrolyzer	Annual FO&M	ϵ_{EL}	66.00	€/(kW year)
Fuel cell	Annual FO&M	ϵ_{HFC}	100.00	€/(kW year)
Gas boiler	Annual FO&M	ϵ_{GB}	1.80	€/(kW year)
Electric boiler	Annual FO&M	ϵ_{EB}	0.80	€/(kW year)
Battery	Annual FO&M	ϵ_{BT}	1.80	€/(kWh year)
H ₂ storage	Annual FO&M	ϵ_{HST}	0.105	€/(kWh year)

Table 4: Retail tariff bounds and bundle discount parameter

Item	Symbol	Value	Unit
Electricity tariff bounds	$[\underline{r}^{\text{elec}}, \bar{r}^{\text{elec}}]$	[0.18, 0.35]	€/kWh
Heat tariff bounds	$[\underline{r}^{\text{heat}}, \bar{r}^{\text{heat}}]$	[0.06, 0.15]	€/kWh
Hydrogen tariff bounds	$[\underline{r}^{\text{hyd}}, \bar{r}^{\text{hyd}}]$	[0.15, 0.45]	€/kWh
Bundle discount factor	α	0.90	–

Table 5: External energy provider tariffs and wholesale prices

Item	Symbol	Value	Unit
External electricity price	$\delta_{mt}^{\text{elec}}$	0.270	€/kWh
External heat price	$\delta_{mt}^{\text{heat}}$	0.100	€/kWh
External hydrogen price	δ_{mt}^{hyd}	0.450	€/kWh
Grid purchase price	$\lambda_{mt}^{\text{Buy}}$	0.1085	€/kWh
Grid selling price	$\lambda_{mt}^{\text{Sell}}$	0.0750	€/kWh
Natural gas tariff	$\beta_{m,t}^{\text{gas}}$	0.0366	€/kWh